

STAGE 1 Factorial Diagnostic: Is the Embedding Ranking Solver-Dependent?

S019A — If the ranking changes when you swap the solver, architecture comparisons are not portable.

SHARED WITH STAGE 2

Same ZCTAs (~32k), same embedding code, same spatial holdout
Same core 3 embeddings: PCA32, Spatial Lag, GraphSAGE
County-holdout evaluation. 5-fold CV. Identical feature pipeline.

VARIED IN STAGE 1

3 solvers (tests whether ranking is portable)

HistGBDT

Ridge

MLP

3 difficulty levels (tests whether effect size is constant)

Easy

Diabetes

Medium

Pop. density

Hard

Elevation

Full factorial: 3 embeddings x 3 solvers x 3 targets = 27 cells x 5 folds = 135 fits
Analysis: two-way ANOVA (embedding x solver) at each difficulty level

What Stage 1 answers

D.2

Is the embedding ranking solver-dependent?

If interaction eta-sq is small, the "best embedding" does not change when you swap solvers. The ranking is portable — architecture comparison is safe.

Gradient

Is sensitivity binary or continuous?

If eta-sq differs across difficulty levels, reporting a single mean effect size misrepresents both easy and hard tasks. Per-regime reporting required.

Stage 1 finding (enables Stage 2)

Interaction eta-sq < 0.04. Embedding eta-sq = 0.55-0.69 (monotonic with difficulty). Ranking is solver-invariant.

Stage 1 validates solver-invariance --> Stage 2 fixes the solver and spends budget on embedding breadth (6 families) and task breadth (27 targets).